

THE TWENTY-SECOND WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

December 2, 1961

Morning Session

1. The graph of the equation $x^y = y^x$ in the first quadrant (i.e., the region where $x > 0$ and $y > 0$) consists of a straight line and a curve. Find the coordinates of the intersection point of the line and the curve. (page 538)

2. For a real-valued function $f(x, y)$ of the two positive real variables x and y , define $f(x, y)$ to be *linearly bounded* if and only if there exists a positive number K such that $|f(x, y)| < (x + y)K$ for all positive x and y . Find necessary and sufficient conditions on the real numbers α and β such that $x^\alpha y^\beta$ is linearly bounded. (page 540)

3. Evaluate

$$\lim_{n \rightarrow \infty} \sum_{j=1}^{n^2} \frac{n}{n^2 + j^2}. \quad (\text{page 541})$$

4. Define a function f over the domain of positive integers as follows: $f(1) = 1$, and for $n > 1$, $f(n) = (-1)^k$ where k is the total number of prime factors of n . For example $f(9) = (-1)^2$, $f(20) = (-1)^3$. Define $F(n)$ as $\sum f(d)$ where the sum ranges over all positive integer divisors of n . Prove that for every positive integer n , $F(n) = 0$ or $F(n) = 1$. For which integers n is $F(n) = 1$? (page 542)

5. Let Ω be a set of n points, where $n > 2$. Let Σ be a nonempty subcollection of the 2^n subsets of Ω that is closed with respect to unions, intersections, and complements (that is, if A and B are members of Σ , then so are $A \cup B$, $A \cap B$, $\Omega - A$ and $\Omega - B$, where $\Omega - B$ denotes all points in Ω but not in B). If k is the number of members of Σ , what are the possible values of k ? Give a proof. (page 543)

6. If $J_2 = \{0, 1\}$ is the field of integers modulo 2, and if $J_2[x]$ is the integral domain of polynomials in one indeterminate with coefficients in J_2 , prove that $p(x) = 1 + x + x^2 + \cdots + x^n$ is reducible (factorable) in case $n + 1$ is composite. Is the converse true? That is, if $n + 1$ is prime, is $p(x)$ irreducible? (page 544)

7. Let S be a nonempty closed set in the Euclidean plane for which there is a closed disk D (a circle together with its interior) containing S such that D is a subset of *every* closed disk that contains S . Prove that every point inside D is the midpoint of a segment joining two points of S . (page 545)

Afternoon Session

1. Let $\alpha_1, \alpha_2, \alpha_3, \dots$ be a sequence of positive real numbers; define s_n as $(\alpha_1 + \alpha_2 + \dots + \alpha_n)/n$ and r_n as $(\alpha_1^{-1} + \alpha_2^{-1} + \dots + \alpha_n^{-1})/n$. Given that $\lim s_n$ and $\lim r_n$ exist as $n \rightarrow \infty$, prove that the product of these limits is not less than 1. (page 546)

2. Let α and β be given positive real numbers, with $\alpha < \beta$. If two points are selected at random from a straight line segment of length β , what is the probability that the distance between them is at least α ? (page 546)

3. Consider four points in a plane, no three of which are collinear, and such that the circle through three of them does not pass through the fourth. Prove that one of the four points can be selected having the property that it lies inside the circle determined by the other three. (page 547)

4. For a fixed positive integer n let $x_1, x_2, \dots, x_n$ be real numbers satisfying $0 \leq x_k \leq 1$ for $k = 1, 2, \dots, n$. Determine the maximum value, as a function of n , of the sum of the $n(n-1)/2$ terms:

$$\sum_{\substack{i,j=1 \\ i < j}}^n |x_i - x_j|. \quad (\text{page 549})$$

5. Let k be a positive integer, and n a positive integer greater than 2. Define

$$f_1(n) = n, f_2(n) = n^{f_1(n)}, \dots, f_{j+1}(n) = n^{f_j(n)}, \text{ etc.}$$

Prove either part of the inequality

$$f_k(n) < n!!! \cdots ! < f_{k+1}(n),$$

where the middle term has k factorial symbols. (page 550)

6. Consider the function $y(x)$ satisfying the differential equation $y'' = -(1 + \sqrt{x})y$ with $y(0) = 1$ and $y'(0) = 0$. Prove that $y(x)$ vanishes exactly once on the interval $0 < x < \pi/2$, and find a positive lower bound for the zero. (page 551)

7. Given a sequence $\{a_n\}$ of non-negative real numbers such that $a_{n+m} \leq a_n a_m$ for all pairs of positive integers, m and n , prove that the sequence $\{\sqrt[n]{a_n}\}$ has a limit as $n \rightarrow \infty$. (page 552)